

Netrokona University
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CSE-1111 Structured Programming Language Lab
Assignment
20 Array Problems

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C PROGRAMMING · LAB ASSIGNMENT

Array Problems

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PROBLEM 01

Highest, Lowest and Average Score

SOURCE CODE

```

#include<stdio.h>
int main(){
    int N;
    float sum = 0;
    scanf("%d", &N);
    int arr[N];
    for(int i=0;i<N;i++){
        scanf("%d",&arr[i]);
    }
    int max=arr[0];
    int min=arr[0];
    for(int i=0;i<N;i++){
        sum = sum + arr[i];
        if(max<=arr[i]){
            max=arr[i];
        }
        if(min>=arr[i]){
            min=arr[i];
        }
    }
    float avg=sum/N;
    printf("Highest = %d\n",max);
    printf("Lowest = %d\n", min);
    printf("Average = %.2f\n",avg);
    return 0;
}

```

OUTPUT

The screenshot shows a C program being executed in a Windows environment. The source code is displayed in a text editor, and the output is shown in a separate window.

Source Code:

```

1  #include<stdio.h>
2  int main() {
3      int N;
4      float sum = 0;
5
6      scanf("%d", &N);
7      int arr[N];
8
9      for(int i=0;i<N;i++){
10         scanf("%d",&arr[i]);
11     }
12
13     int max=arr[0];
14     int min=arr[0];
15
16     for(int i=0;i<N;i++){
17         sum = sum + arr[i];
18
19         if(max<=arr[i]){
20             max=arr[i];
21         }
22
23         if(min>=arr[i]){
24             min=arr[i];
25         }
26     }
27
28     float avg=sum/N;
29
30     printf("Highest = %d\n",max);
31     printf("Lowest = %d\n", min);
32     printf("Average = %.2f\n",avg);
33
34     return 0;
35 }
36

```

Output:

```

6
72 85 64 91 78 69
Highest = 91
Lowest = 64
Average = 76.50

Process returned 0 (0x0)
execution time : 42.858 s
Press any key to continue.

```

PROBLEM 02

Maximum and Minimum Temperature

SOURCE CODE

```

#include<stdio.h>
int main(){
    int N;
    scanf("%d",&N);
    int arr[N];
    for(int i=0;i<N;i++){
        scanf("%d",&arr[i]);
    }
    int max=arr[0];
    int min=arr[0];
    for(int i=0;i<N;i++){
        if(max<=arr[i]){
            max=arr[i];
        }
        if(min>=arr[i]){
            min=arr[i];
        }
    }
    printf("Maximum Temperature = %d\n",max);
    printf("Minimum Temperature = %d\n",min);
    return 0;
}

```

OUTPUT

The screenshot shows a C code editor with the following code:

```

1  #include<stdio.h>
2  int main() {
3      int N;
4
5      scanf("%d", &N);
6      int arr[N];
7
8      for(int i=0; i<N; i++) {
9          scanf("%d", &arr[i]);
10     }
11
12     int max=arr[0];
13     int min=arr[0];
14
15     for(int i=0; i<N; i++) {
16         if(max<=arr[i]) {
17             max=arr[i];
18         }
19         if(min>=arr[i]) {
20             min=arr[i];
21         }
22     }
23
24     printf("Maximum Temperature = %d\n", max);
25     printf("Minimum Temperature = %d\n", min);
26
27     return 0;
28 }

```

The output window shows the following text:

```

7
31 34 29 36 33 28 35
Maximum Temperature = 36
Minimum Temperature = 28

Process returned 0 (0x0)
execution time : 45.020 s
Press any key to continue.

```

PROBLEM 03

Total and Average

SOURCE CODE

```

#include<stdio.h>
int main(){
    int N;
    int sum=0;
    scanf("%d",&N);
    int arr[N];

    for(int i=0;i<N;i++){
        scanf("%d",&arr[i]);
        sum = sum + arr[i];
    }
    float avg = (float)sum/N;

    printf("Total = %d\n",sum);
    printf("Average = %.2f\n",avg);
}

```

OUTPUT

The screenshot shows a C code editor with the following code:

```

1  #include<stdio.h>
2  int main(){
3      int N;
4      int sum=0;
5      scanf("%d",&N);
6      int arr[N];
7
8      for(int i=0;i<N;i++){
9          scanf("%d",&arr[i]);
10         sum = sum + arr[i];
11     }
12     float avg = (float)sum/N;
13
14     printf("Total = %d\n",sum);
15     printf("Average = %.2f\n",avg);
16 }
17

```

The output window shows the following results:

```

5
12 15 10 18 20
Total = 75
Average = 15.00

Process returned 0 (0x0)
execution time : 33.195 s
Press any key to continue.

```

PROBLEM 04

Days Above Target

SOURCE CODE

```
#include<stdio.h>
int main()
{
    int N,t,count=0;
    scanf("%d",&N);

    int arr[N];

    for(int i=0; i<N; i++)
    {
        scanf("%d",&arr[i]);
    }

    scanf("%d",&t);

    for(int i=0;i<N;i++){
        if(t<arr[i]){
            count++;
        }
    }

    printf("Days above target = %d",count);

    return 0;
}
```

OUTPUT

The screenshot shows a C code editor window titled "4.c - Code::Blocks 25.03". The code is the same as provided in the source code section. To the right of the editor, a terminal window titled "F:\Shimu\Shuvon Sir Array" displays the output of the program. The output shows the input values 6, 25, 40, 18, 55, 32, 47, and 30, followed by the message "Days above target = 4", "Process returned 0 (0x0)", "execution time : 40.765 s", and "Press any key to continue.".

```
1  #include<stdio.h>
2  int main()
3  {
4      int N,t,count=0;
5      scanf("%d",&N);
6
7      int arr[N];
8
9      for(int i=0; i<N; i++)
10     {
11         scanf("%d",&arr[i]);
12     }
13
14     scanf("%d",&t);
15
16     for(int i=0;i<N;i++){
17         if(t<arr[i]){
18             count++;
19         }
20     }
21
22     printf("Days above target = %d",count);
23
24     return 0;
25 }
26
```

```
6
25 40 18 55 32 47
30
Days above target = 4
Process returned 0 (0x0)
execution time : 40.765 s
Press any key to continue.
```

PROBLEM 05

Pass and Fail Count

SOURCE CODE

```

#include<stdio.h>
int main(){
    int N,pass=0,fail=0;
    scanf("%d",&N);

    int arr[N];
    for(int i=0;i<N;i++){
        scanf("%d",&arr[i]);
        if(arr[i]>=40) pass++;
        else fail++;
    }

    printf("Passed = %d\n",pass);
    printf("Failed = %d\n",fail);

    return 0;
}

```

OUTPUT

```

5.c - Code::Blocks 25.03
File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help
Start here 1.c 2.c 3.c 4.c 5.c 6.c 7.c 8.c 9.c 10.c 11.c 12.c 13.c 14.c 15.c 16.c 17.c 18.c 19.c 20.c
1  #include<stdio.h>
2  int main(){
3      int N,pass=0,fail=0;
4      scanf("%d",&N);
5
6      int arr[N];
7      for(int i=0;i<N;i++){
8          scanf("%d",&arr[i]);
9          if(arr[i]>=40) pass++;
10         else fail++;
11     }
12
13     printf("Passed = %d\n",pass);
14     printf("Failed = %d\n",fail);
15
16     return 0;
17 }
18
F:\Shimu\Shuvon Sir Array\5.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 1, Col 1, Pos 0 Insert

```

```

8
35 72 41 28 56 39 80 44
Passed = 5
Failed = 3

Process returned 0 (0x0)
execution time : 30.482 s
Press any key to continue.

```

PROBLEM 06

Highest and Lowest Price

SOURCE CODE

```

#include<stdio.h>
int main(){
    int N;

    scanf("%d",&N);
    int arr[N];

    for(int i=0;i<N;i++){
        scanf("%d",&arr[i]);
    }

    int max=arr[0];
    int min=arr[0];

    for(int i=0;i<N;i++){
        if(max<=arr[i]){
            max=arr[i];
        }
        if(min>=arr[i]){
            min=arr[i];
        }
    }

    printf("Highest Price = %d\n",max);
    printf("Lowest Price = %d\n",min);

    return 0;
}

```

OUTPUT

The screenshot displays a C code editor with the following code:

```

1  #include<stdio.h>
2  int main() {
3      int N;
4
5      scanf("%d",&N);
6      int arr[N];
7
8      for(int i=0;i<N;i++){
9          scanf("%d",&arr[i]);
10     }
11
12     int max=arr[0];
13     int min=arr[0];
14
15     for(int i=0;i<N;i++){
16         if(max<=arr[i]){
17             max=arr[i];
18         }
19         if(min>=arr[i]){
20             min=arr[i];
21         }
22     }
23
24     printf("Highest Price = %d\n",max);
25     printf("Lowest Price = %d\n",min);
26
27     return 0;
28 }
29

```

The output window shows the following execution results:

```

5
450 275 800 320 650
Highest Price = 800
Lowest Price = 275

Process returned 0 (0x0)
execution time : 33.373 s
Press any key to continue.

```

PROBLEM 07

Total and Average Salary

SOURCE CODE

```
#include<stdio.h>
int main(){
    int N;
    int sum=0;
    scanf("%d",&N);
    int arr[N];

    for(int i=0;i<N;i++){
        scanf("%d",&arr[i]);
        sum = sum + arr[i];
    }
    float avg = (float)sum/N;

    printf("Total Salary = %d\n",sum);
    printf("Average Salary = %.2f\n",avg);
}
```

OUTPUT

The screenshot shows a C++ IDE with the source code for 'Total and Average Salary' on the left and its output on the right. The code is identical to the one provided in the 'SOURCE CODE' section. The output window shows the program's execution results.

```
5
30000 35000 28000 42000 32000
Total Salary = 167000
Average Salary = 33400.00

Process returned 0 (0x0)   exe
cution time : 41.421 s
Press any key to continue.
```

PROBLEM 08

Count Even and Odd Numbers

SOURCE CODE

```

#include <stdio.h>
int main()
{
    int N, even = 0, odd = 0;

    scanf("%d", &N);

    int arr[N];

    for (int i = 0; i < N; i++)
    {
        scanf("%d", &arr[i]);

        if (arr[i] % 2 == 0)
        {
            even++;
        }
        else
        {
            odd++;
        }
    }

    printf("Even = %d\n", even);
    printf("Odd = %d\n", odd);
}

```

OUTPUT

The screenshot shows a C code editor with the following code:

```

1  #include <stdio.h>
2  int main()
3  {
4      int N, even = 0, odd = 0;
5
6      scanf("%d", &N);
7
8      int arr[N];
9
10     for (int i = 0; i < N; i++)
11     {
12         scanf("%d", &arr[i]);
13
14         if (arr[i] % 2 == 0)
15         {
16             even++;
17         }
18         else
19         {
20             odd++;
21         }
22     }
23
24     printf("Even = %d\n", even);
25     printf("Odd = %d\n", odd);
26 }

```

The output window shows the following text:

```

8
12 7 15 20 18 9 4 11
Even = 4
Odd = 4

Process returned 0 (0x0)
execution time : 31.022 s
Press any key to continue.

```

PROBLEM 09

Count Deposits and Withdrawals

SOURCE CODE

```

#include<stdio.h>
int main(){
    int N,Withdrawals=0,Deposits=0;

    scanf("%d",&N);
    int arr[N];

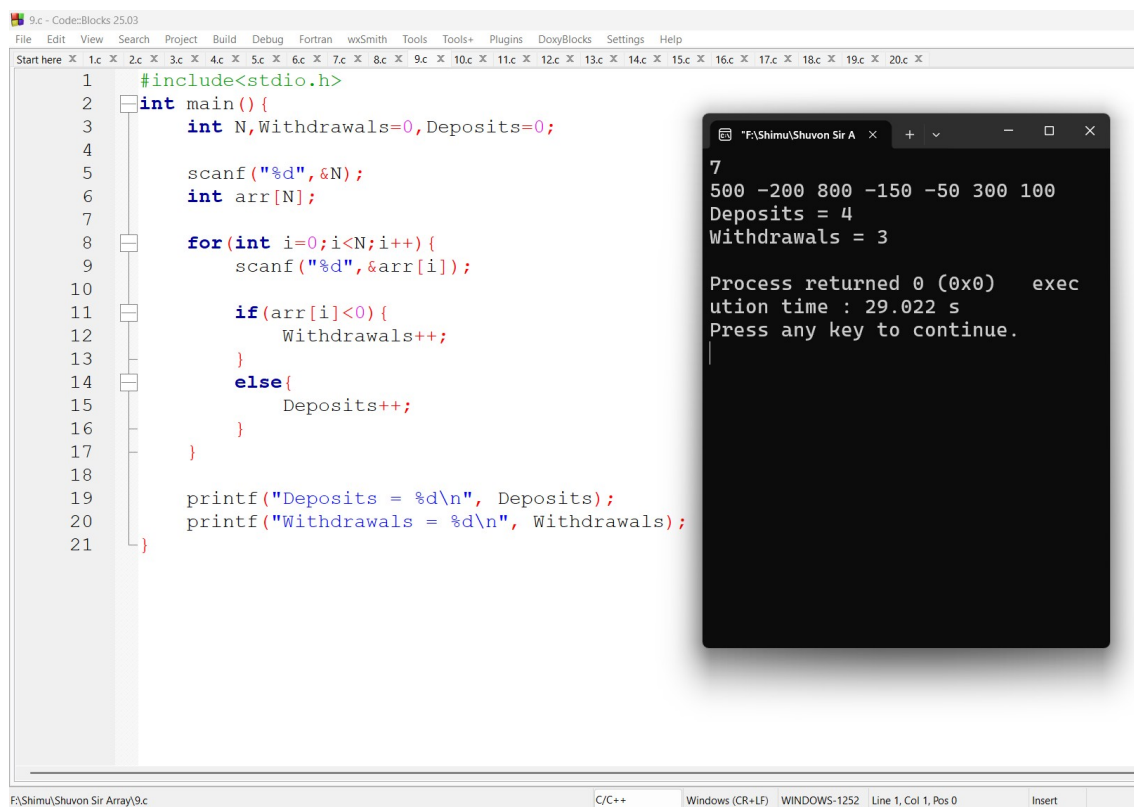
    for(int i=0;i<N;i++){
        scanf("%d",&arr[i]);

        if(arr[i]<0){
            Withdrawals++;
        }
        else{
            Deposits++;
        }
    }

    printf("Deposits = %d\n", Deposits);
    printf("Withdrawals = %d\n", Withdrawals);
}

```

OUTPUT



The screenshot displays a C code editor with the following source code:

```

1  #include<stdio.h>
2  int main() {
3      int N,Withdrawals=0,Deposits=0;
4
5      scanf("%d",&N);
6      int arr[N];
7
8      for(int i=0;i<N;i++){
9          scanf("%d",&arr[i]);
10
11         if(arr[i]<0){
12             Withdrawals++;
13         }
14         else{
15             Deposits++;
16         }
17     }
18
19     printf("Deposits = %d\n", Deposits);
20     printf("Withdrawals = %d\n", Withdrawals);
21 }

```

The output window shows the following results:

```

7
500 -200 800 -150 -50 300 100
Deposits = 4
Withdrawals = 3

Process returned 0 (0x0)   exec
ution time : 29.022 s
Press any key to continue.

```

PROBLEM 10

Average of Values Meeting a Threshold

SOURCE CODE

```

#include<stdio.h>
int main(){
    int N,threshold,count=0,sum=0;
    scanf("%d",&N);
    int arr[N];

    for(int i=0;i<N;i++){
        scanf("%d",&arr[i]);
    }
    scanf("%d",&threshold);

    for(int i=0;i<N;i++){
        if(arr[i]>=threshold){
            count++;
        }
        else{
            arr[i]=0;
        }
    }
    for(int i=0;i<N;i++){
        sum+=arr[i];
    }
    float average=(float)sum/count;
    printf("Average = %.2f\n", average);
    return 0;
}

```

OUTPUT

The screenshot displays a C code editor with the following code and its output:

```

1  #include<stdio.h>
2  int main() {
3      int N,threshold,count=0,sum=0;
4      scanf("%d",&N);
5      int arr[N];
6
7      for(int i=0;i<N;i++){
8          scanf("%d",&arr[i]);
9      }
10     scanf("%d",&threshold);
11
12     for(int i=0;i<N;i++){
13         if(arr[i]>=threshold){
14             count++;
15         }
16         else{
17             arr[i]=0;
18         }
19     }
20     for(int i=0;i<N;i++){
21         sum+=arr[i];
22     }
23     float average=(float)sum/count;
24     printf("Average = %.2f\n", average);
25     return 0;
26 }

```

The output window shows the following text:

```

6
35 55 72 40 29 81
40
Average = 62.00

Process returned 0 (0x0)
execution time : 47.538 s
Press any key to continue.

```

PROBLEM 11

Move Last Element to the Front

SOURCE CODE

```
#include<stdio.h>
int main(){
    int N;

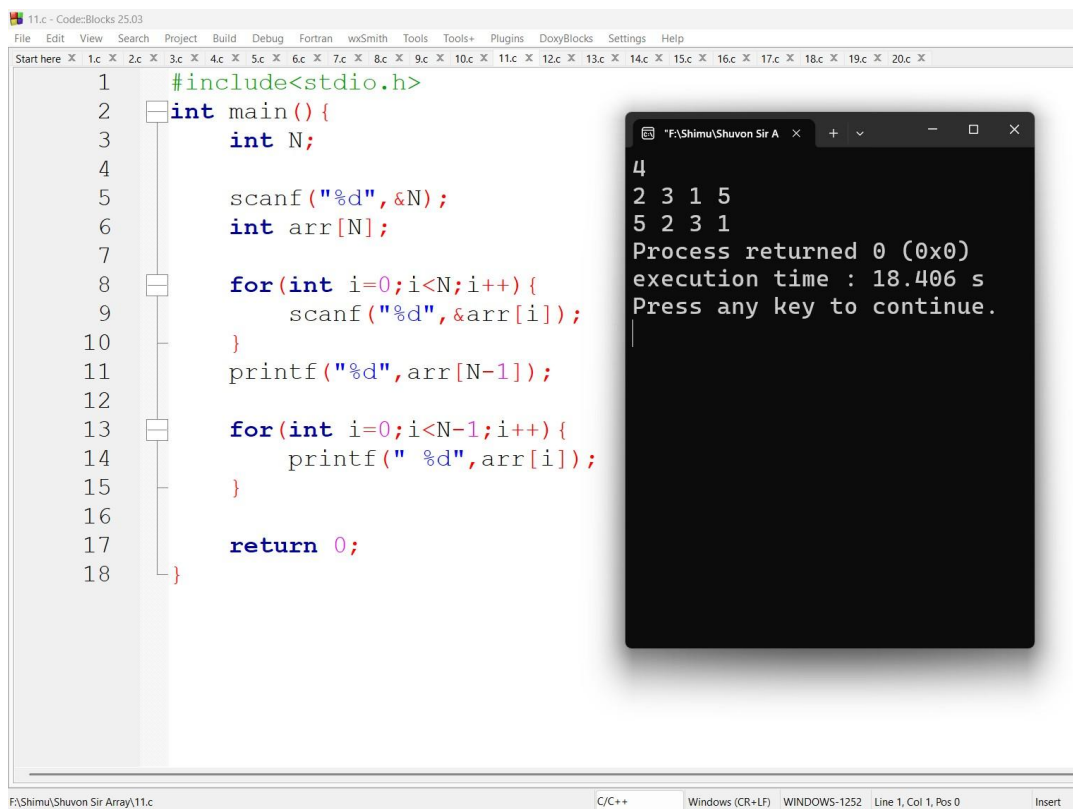
    scanf("%d",&N);
    int arr[N];

    for(int i=0;i<N;i++){
        scanf("%d",&arr[i]);
    }
    printf("%d",arr[N-1]);

    for(int i=0;i<N-1;i++){
        printf(" %d",arr[i]);
    }

    return 0;
}
```

OUTPUT



The screenshot shows a C code editor window titled "11.c - Code::Blocks 25.03". The code is the same as provided in the source code block. To the right of the editor is a terminal window titled "F:\Shimu\Shuvon Sir A". The terminal output shows the number 4, followed by the array elements 2 3 1 5 on the next line, and 5 2 3 1 on the line after. Below the array elements, the terminal displays "Process returned 0 (0x0)", "execution time : 18.406 s", and "Press any key to continue.".

```
11.c - Code::Blocks 25.03
File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help
Start here x 1.c x 2.c x 3.c x 4.c x 5.c x 6.c x 7.c x 8.c x 9.c x 10.c x 11.c x 12.c x 13.c x 14.c x 15.c x 16.c x 17.c x 18.c x 19.c x 20.c x
1 #include<stdio.h>
2 int main() {
3     int N;
4
5     scanf("%d", &N);
6     int arr[N];
7
8     for(int i=0; i<N; i++) {
9         scanf("%d", &arr[i]);
10    }
11    printf("%d", arr[N-1]);
12
13    for(int i=0; i<N-1; i++) {
14        printf(" %d", arr[i]);
15    }
16
17    return 0;
18 }
F:\Shimu\Shuvon Sir A
4
2 3 1 5
5 2 3 1
Process returned 0 (0x0)
execution time : 18.406 s
Press any key to continue.
```

PROBLEM 12

Move First Element to the End

SOURCE CODE

```
#include<stdio.h>
int main(){
    int N;
    scanf("%d",&N);
    int arr[N];

    for (int i=0;i<N;i++){
        scanf("%d",&arr[i]);
    }

    for(int i=1;i<N;i++){
        printf("%d ",arr[i]);
    }
    printf("%d",arr[0]);

    return 0;
}
```

OUTPUT

The screenshot shows a C code editor with the following code:

```
1  #include<stdio.h>
2  int main() {
3      int N;
4      scanf("%d",&N);
5      int arr[N];
6
7      for (int i=0;i<N;i++) {
8          scanf("%d",&arr[i]);
9      }
10
11     for(int i=1;i<N;i++) {
12         printf("%d ",arr[i]);
13     }
14     printf("%d",arr[0]);
15
16     return 0;
17 }
```

The output window shows the following text:

```
5
10 20 30 40 50
20 30 40 50 10
Process returned 0 (0x0)
execution time : 19.500 s
Press any key to continue.
```

PROBLEM 13

Print Array in Reverse Order

SOURCE CODE

```
#include<stdio.h>
int main(){
    int N;
    scanf("%d",&N);
    int arr[N];

    for(int i=0;i<N;i++){
        scanf("%d",&arr[i]);
    }

    for(int i=N-1;i>=0;i--){
        printf("%d ",arr[i]);
    }
    return 0;
}
```

OUTPUT

The screenshot displays a C code editor with the following code:

```
1 #include<stdio.h>
2 int main(){
3     int N;
4     scanf("%d",&N);
5     int arr[N];
6
7     for(int i=0;i<N;i++){
8         scanf("%d",&arr[i]);
9     }
10
11     for(int i=N-1;i>=0;i--){
12         printf("%d ",arr[i]);
13     }
14     return 0;
15 }
```

The output window shows the execution results:

```
6
12 15 18 21 24 27
27 24 21 18 15 12
Process returned 0 (0x0)
execution time : 31.212 s
Press any key to continue.
```

PROBLEM 14

Move All Zeros to the End

SOURCE CODE

```
#include<stdio.h>
int main(){
    int N,count=0;
    scanf("%d",&N);
    int arr[N];

    for(int i=0;i<N;i++){
        scanf("%d",&arr[i]);
        if(arr[i]==0){
            count++;
        }
    }

    for(int i=0;i<N;i++){
        if(arr[i]!=0){
            printf("%d ",arr[i]);
        }
    }
    for(int i=0;i<count;i++){
        printf("0 ");
    }
    return 0;
}
```

OUTPUT

The screenshot shows a code editor with the following C code:

```
1 #include<stdio.h>
2 int main(){
3     int N,count=0;
4     scanf("%d",&N);
5     int arr[N];
6
7     for(int i=0;i<N;i++){
8         scanf("%d",&arr[i]);
9         if(arr[i]==0){
10             count++;
11         }
12     }
13
14     for(int i=0;i<N;i++){
15         if(arr[i]!=0){
16             printf("%d ",arr[i]);
17         }
18     }
19     for(int i=0;i<count;i++){
20         printf("0 ");
21     }
22     return 0;
23 }
```

The output window shows the following text:

```
8
4 0 7 0 2 5 0 9
4 7 2 5 9 0 0 0
Process returned 0 (0x0)
execution time : 33.360 s
Press any key to continue.
```

PROBLEM 15

Remove Duplicate Elements

SOURCE CODE

```

#include<stdio.h>
int main(){
    int N;
    scanf("%d",&N);
    int arr[N];

    for(int i=0;i<N;i++){
        scanf("%d",&arr[i]);
    }
    for(int i=0;i<N;i++){
        for(int j=i+1;j<N;j++){
            if(arr[i]==arr[j]){
                arr[j]=0;
            }
        }
    }

    for(int i=0;i<N;i++){
        if(arr[i]!=0){
            printf("%d ",arr[i]);
        }
    }

    return 0;
}

```

OUTPUT

The screenshot shows a C code editor with the following code:

```

1  #include<stdio.h>
2  int main() {
3      int N;
4      scanf("%d", &N);
5      int arr[N];
6
7      for(int i=0; i<N; i++){
8          scanf("%d", &arr[i]);
9      }
10     for(int i=0; i<N; i++){
11         for(int j=i+1; j<N; j++){
12             if(arr[i]==arr[j]){
13                 arr[j]=0;
14             }
15         }
16     }
17
18     for(int i=0; i<N; i++){
19         if(arr[i]!=0){
20             printf("%d ", arr[i]);
21         }
22     }
23
24     return 0;
25 }

```

The output window shows the following text:

```

9
101 103 101 105 107 103 109 105 110
101 103 105 107 109 110
Process returned 0 (0x0)   execution
time : 60.752 s
Press any key to continue.

```

PROBLEM 16

Print Duplicate Elements

SOURCE CODE

```

#include<stdio.h>
int main(){
    int N;
    scanf("%d",&N);
    int arr[N];

    for(int i=0;i<N;i++){
        scanf("%d",&arr[i]);
    }
    for(int i=0;i<N;i++){
        for(int j=i+1;j<N;j++){
            if(arr[i]==arr[j]){
                printf("%d ",arr[i]);
            }
        }
    }

    return 0;
}

```

OUTPUT

The screenshot shows a code editor with the following C code:

```

1  #include<stdio.h>
2  int main(){
3      int N;
4      scanf("%d",&N);
5      int arr[N];
6
7      for(int i=0;i<N;i++){
8          scanf("%d",&arr[i]);
9      }
10     for(int i=0;i<N;i++){
11         for(int j=i+1;j<N;j++){
12             if(arr[i]==arr[j]){
13                 printf("%d ",arr[i]);
14             }
15         }
16     }
17
18     return 0;
19 }
20

```

The output window shows the following text:

```

9
12 15 18 12 20 15 25 18 30
12 15 18
Process returned 0 (0x0)
execution time : 36.614 s
Press any key to continue.

```

PROBLEM 17

Print Unique Elements

SOURCE CODE

```
#include<stdio.h>
int main(){
    int N;
    scanf("%d",&N);
    int arr[N];

    for(int i=0;i<N;i++){
        scanf("%d",&arr[i]);
    }
    for(int i=0;i<N;i++){
        int freq=0;
        for(int j=0;j<N;j++){
            if(arr[i]==arr[j]){
                freq++;
            }
        }
        if(freq==1){
            printf("%d ",arr[i]);
        }
    }
    return 0;
}
```

OUTPUT

The screenshot shows a C code editor with the following code:

```
1 #include<stdio.h>
2 int main() {
3     int N;
4     scanf("%d", &N);
5     int arr[N];
6
7     for(int i=0; i<N; i++) {
8         scanf("%d", &arr[i]);
9     }
10    for(int i=0; i<N; i++) {
11        int freq=0;
12        for(int j=0; j<N; j++) {
13            if(arr[i]==arr[j]) {
14                freq++;
15            }
16        }
17        if(freq==1) {
18            printf("%d ", arr[i]);
19        }
20    }
21    return 0;
22 }
```

The output of the program is shown in a terminal window:

```
9
1 2 2 4 6 3 1 2 4
6 3
Process returned 0 (0x0)
execution time : 38.225 s
Press any key to continue.
```

PROBLEM 18

Second Highest Element

SOURCE CODE

```

#include<stdio.h>
int main(){
    int N, max=0;
    scanf("%d",&N);
    int arr[N];
    for(int i=0;i<N;i++){
        scanf("%d",&arr[i]);
    }

    for(int i=0;i<N;i++){
        int second_max=arr[i];
        for(int j=i+1;j<N;j++){
            if(arr[j]>second_max){
                second_max=arr[j];
                max = second_max;
            }
        }
    }
    printf("Second Highest = %d ", max);
    return 0;
}

```

OUTPUT

```

1  #include<stdio.h>
2  int main() {
3      int N, max=0;
4      scanf("%d", &N);
5      int arr[N];
6      for (int i=0; i<N; i++) {
7          scanf("%d", &arr[i]);
8      }
9
10     for (int i=0; i<N; i++) {
11         int second_max=arr[i];
12         for (int j=i+1; j<N; j++) {
13             if (arr[j]>second_max) {
14                 second_max=arr[j];
15                 max = second_max;
16             }
17         }
18     }
19     printf("Second Highest = %d ", max);
20     return 0;
21 }
22

```

Output:

```

7
72 91 85 91 64 88 79
Second Highest = 88
Process returned 0 (0x0)
execution time : 40.703 s
Press any key to continue.

```

PROBLEM 19

Second Lowest Element

SOURCE CODE

```

#include<stdio.h>
#include<limits.h>
int main(){
    int N;
    scanf("%d",&N);
    int arr[N];
    for(int i=0;i<N;i++){
        scanf("%d",&arr[i]);
    }
    int first=INT_MAX,second=INT_MAX;
    for(int i=0;i<N;i++){
        if(arr[i]<first){
            second=first;
            first=arr[i];
        }
        else if(arr[i]<second && arr[i]!=first){
            second=arr[i];
        }
    }
    printf("Second Lowest = %d\n", second);

    return 0;
}

```

OUTPUT

The screenshot shows a code editor with the C program code on the left and a terminal window on the right. The code is the same as provided in the 'SOURCE CODE' section. The terminal window shows the output of the program when executed with the input 7 and the array [500, 250, 350, 250, 700, 400, 600]. The output is 'Second Lowest = 350'. Below the output, the terminal shows 'Process returned 0 (0x0)', 'execution time : 38.760 s', and 'Press any key to continue.'.

```

1  #include<stdio.h>
2  #include<limits.h>
3  int main(){
4      int N;
5      scanf("%d",&N);
6      int arr[N];
7      for(int i=0;i<N;i++){
8          scanf("%d",&arr[i]);
9      }
10     int first=INT_MAX,second=INT_MAX;
11     for(int i=0;i<N;i++){
12         if(arr[i]<first){
13             second=first;
14             first=arr[i];
15         }
16         else if(arr[i]<second && arr[i]!=first){
17             second=arr[i];
18         }
19     }
20     printf("Second Lowest = %d\n", second);
21
22     return 0;
23 }
24
25

```

```

7
500 250 350 250 700 400 600
Second Lowest = 350

Process returned 0 (0x0)
execution time : 38.760 s
Press any key to continue.

```

PROBLEM 20

Find the Missing Number

SOURCE CODE

```
#include <stdio.h>
#define MAX 10000

int main() {
    int arr[MAX], n, i;
    long long expected, actual = 0;

    scanf("%d", &n);

    for (i = 0; i < n - 1; i++) {
        scanf("%d", &arr[i]);
        actual += arr[i];
    }

    expected = (long long)n * (n + 1) / 2;
    printf("Missing Number = %lld\n", expected - actual);
    return 0;
}
```

OUTPUT

The screenshot shows a C code editor with the source code for finding a missing number. The code is as follows:

```
#include <stdio.h>
#define MAX 10000

int main() {
    int arr[MAX], n, i;
    long long expected, actual = 0;

    scanf("%d", &n);

    for (i = 0; i < n - 1; i++) {
        scanf("%d", &arr[i]);
        actual += arr[i];
    }

    expected = (long long)n * (n + 1) / 2;
    printf("Missing Number = %lld\n", expected - actual);
    return 0;
}
```

The output window shows the following text:

```
6
1 2 3 5 6
Missing Number = 4

Process returned 0 (0x0)
execution time : 29.823 s
Press any key to continue.
```